SIMPLEX composition-space deep learning for hydrogel adhesion screening]SIMPLEX Composition-Space Deep Learning for Bio-inspired Hydrogel Adhesion with Prospective Validation on Model-Discovered Formulations and Interpretable Composition Rules

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Abstract

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The design of functional hydrogels for biohybrid interfaces, wet load-bearing junctions between living tissue and synthetic materials, relies on trial-and-error experimentation, because every candidate formulation must be synthesised and mechanically characterised before its underwater adhesion can be assessed. A data-driven model that predicts adhesion strength from monomer composition can accelerate candidate screening in this data-scarce regime. Here we introduce SIMPLEX, a dual-modality residual network with interaction self-attention and gated fusion that maps six monomer molar fractions, together with their 15 pairwise interaction terms that encode composition synergy explicitly, to underwater glass adhesion strength (kPa), trained under Mixup, stochastic weight averaging and a range-domain constraint. We train on 316 hydrogel formulations spanning the full experimentally explored adhesion range (1 to 353 kPa; 19% above 150 kPa) and prospectively validate on 25 held-out formulations discovered in the final round of the original study's sequential model-based optimisation loop (adhesion 62 to 251 kPa), a held-out model-guided cohort that was never touched during model selection. Across 5-fold grouped cross-validation repeated over 10 seeds, SIMPLEX reaches $R^2 = 0.79$, matching the strongest tree baseline (random forest, 0.81); the difference is within statistical noise (Holm-corrected $p = 1.0$). On the 25-formulation prospective cohort, SIMPLEX attains the highest prospective accuracy of the equally tuned baselines ($R^2 = 0.69$, versus 0.63 for the strongest baseline SVR-RBF, a gain of +0.060), together with the highest ranking correlation (Spearman $\rho = 0.87$) and perfect Top-10 recovery (1.00); the internal-to-prospective generalisation gap of SIMPLEX (0.79 to 0.69) is roughly half that of random forest (0.81 to 0.56). Ablation identifies multimodal fusion as the largest single contribution, followed by the interaction self-attention and Mixup regularisation, and permutation importance highlights the cationic monomer and pairwise interaction terms, revealing the hydrophobic-aromatic BA×PEA interaction as the dominant driver of underwater adhesion. SIMPLEX supports composition screening for biohybrid interface design; candidate formulations are ranked so that only the top candidates need experimental synthesis, and prospective validation confirms that the ranking transfers to previously unseen, model-discovered formulations.

2 Keywords

Hydrogel, Biohybrid interfaces, Machine learning, Composition-to-property prediction, Prospective validation, Material screening, Adhesion strength

3 Introduction

Hydrogel design currently relies on trial-and-error experimentation, where each formulation must be synthesised and mechanically characterised ???????. A data-driven model that predicts adhesion strength from monomer composition can dramatically accelerate screening of candidate formulations for biomedical and engineering applications, in particular at biohybrid interfaces where wet, load-bearing junctions between living tissue and synthetic materials demand carefully tuned adhesion ???????????. The practical value of such predictions is direct; they narrow the experimental search space, prioritise which samples deserve costly follow-up, and expose which measured quantities actually carry information ?.

Three families of methods dominate current practice. Classical regularised linear models and kernel methods are stable at small sample sizes but cannot express the interactions between modalities that domain experts know to exist ????????. Tree ensembles (random forests and gradient boosting) are the de-facto standard on tabular data, but they treat every column as an exchangeable scalar and provide no mechanism for sharing statistical strength across related endpoints ??. Deep tabular networks promise both, yet in this regime they usually underperform; with a few hundred samples they overfit ?????.

The failure modes are specific rather than generic. First, evaluation protocols that split rows at random allow measurements from the same source to appear in both training and test partitions, which inflates reported accuracy ??. Second, external validation on a cohort collected independently is rarely attempted, so it remains unknown whether learned relationships transfer at all ????. Third, and most damaging for soft-materials machine learning, when a held-out cohort does exist it is usually evaluated after model selection has already exploited the full dataset, so the reported external numbers partly measure effort rather than generalisation ????.

Here we present **SIMPLEX** (Simplex composition encoding with Interaction-aware attention, Multi-modal fusion, Pretraining-ready regularisation, Learnable domain constraints and EXtrapolation evaluation; the extrapolation is in composition space, while the target range of the prospective cohort remains interpolative with respect to training). SIMPLEX encodes the composition through two explicit modalities, the raw monomer fractions and their pairwise interactions, so that the model can represent composition synergy rather than relying on the network to invent it; refines the fused representation with residual blocks separated by an interaction self-attention layer and a gated fusion mechanism; and predicts adhesion strength under Mixup, stochastic weight averaging and a range-domain constraint **??????????????????????**. Every design choice is an ablation switch, and any switch that does not pay for itself is removed from the final configuration; we also evaluated a simplex-projected Mixup variant (SimplexMix) and report it as non-beneficial, in line with the ablation-gating principle **????**.

Our contributions are as follows. (i) a fully public, registration-free benchmark of 316 training and 25 prospectively held-out model-guided formulations with 21 features across two modalities; (ii) SIMPLEX, a dual-modality residual architecture with interaction attention and small-data regularisation; (iii) an evaluation under repeated grouped cross-validation against equally tuned baselines, with a single, frozen-model pass on the 25-formulation prospective cohort; and (iv) evidence that SIMPLEX transfers to previously unseen, model-discovered formulations with the highest prospective accuracy of the equally tuned baselines ($R^2 = 0.69$, versus 0.63 for the strongest baseline SVR-RBF) and the highest ranking performance (Spearman 0.87, Top-10 recovery 1.00), supporting composition screening for biohybrid interface design; and (v) interpretable composition rules (the hydrophobic-aromatic BA×PEA interaction as the dominant adhesion driver, with a monotone partial-dependence design rule) for guiding formulation design on the composition simplex.

4 Materials and Methods

4.1 Overall workflow and design rationale

The workflow is summarised in (figpipeline). Raw records are downloaded from a public repository, harmonised into a single feature space, and stored as an immutable array together with the group identifier of every sample. Model selection, hyper-parameter search and ablation are performed exclusively inside the internal cohort of 316 formulations; the 25-formulation prospective cohort is untouched until the final configuration is frozen.

4.2 Data acquisition

4.2.1 Internal dataset

The internal cohort comprises 316 bio-inspired formulations from the publicly released dataset *shenghu/hydrogels* (MIT licence) accompanying Liao et al., *Nature* 644, 89 to 95 (2025) **?**. Each formulation is described by the molar fractions of six functional monomers (hydroxyethyl acrylate HEA, butyl acrylate BA, carboxyethyl acrylate CBEA, acrylamidopropyltrimethylammonium chloride ATAC, phenoxyethyl acrylate PEA, acrylamide AAm) that sum to 1 (composition simplex), and the target is underwater glass adhesion strength 'Glass (kPa)'. The internal adhesion ranges from 1 to 353 kPa (mean 93 kPa), with 19% of samples above 150 kPa, so the model is trained on the full experimentally explored composition-adhesion surface.

4.2.2 Prospective validation cohort

The prospective cohort comprises the 25 formulations added in the final round of the original study's sequential model-based optimisation (SMBO) loop **?**. These are the last model-guided discoveries (adhesion 62 to 251 kPa, mean 158 kPa) and were not used in any way for model selection, tuning, or ablation **?**; they are a genuinely held-out, prospective cohort; both cohorts are characterised in (fig:dataset).

4.2.3 Features

Two modalities are constructed (21 features), following compositional-data and materials-informatics practice **???**; modality 1 is the six raw monomer fractions; modality 2 is the 15 pairwise interaction terms $x_i x_j$

$(i < j)$, encoding monomer synergy explicitly. Every formulation is a point on the unit simplex

$$\mathcal{S}^5 = \left\{ \mathbf{x} \in \mathbb{R}^6_{\geq 0} \; : \; \sum_{k=1}^{6} x_k = 1 \right\}, \tag{1}$$

so the interaction modality $\mathbf{z} = \{x_i x_j\}_{i<j}$ is fully determined by $\mathbf{x}$ and adds no degrees of freedom, only an explicit, learnable representation of composition synergy. Scaling and any imputation are fitted inside each CV training fold only (no leakage).

4.3 Architecture of SIMPLEX

The architecture is detailed in (figmodelarch). The two modalities are embedded to a common dimension ($d = 152$), refined by two residual blocks separated by an interaction self-attention layer (8 heads), combined by a gated fusion mechanism, pooled, and mapped by a linear head to a non-negative adhesion prediction (kPa) under a range-domain constraint. The gated fusion combines the two modality embeddings $\mathbf{h}_1, \mathbf{h}_2$ as

$$\mathbf{g} = \sigma\big(\mathbf{W}_g[\mathbf{h}_1; \mathbf{h}_2] + \mathbf{b}_g\big), \qquad \mathbf{h} = \mathbf{g} \odot \mathbf{h}_1 + (1 - \mathbf{g}) \odot \mathbf{h}_2, \tag{2}$$

where σ is the logistic sigmoid, $\odot$ the elementwise product, and $[\cdot;\cdot]$ concatenation; the gate learns per-dimension routing between the raw-fraction and interaction modalities. The gated fusion strategy was selected over concatenation, cross-attention, and FiLM-style conditioning in the internal ablation-gated search. Training uses Mixup ($\alpha = 0.4$), stochastic weight averaging, early stopping on an inner validation split, and a constraint penalty on out-of-range predictions. Mixup augments the input space by convex combination of training pairs,

$$\tilde{\mathbf{x}} = \lambda \mathbf{x}_a + (1 - \lambda)\mathbf{x}_b, \qquad \tilde{y} = \lambda y_a + (1 - \lambda) y_b, \qquad \lambda \sim \mathrm{Beta}(\alpha, \alpha), \tag{3}$$

which acts as a simplex-compatible regulariser for composition-to-property learning. The domain constraint penalises predictions that violate the physically admissible adhesion range,

$$\mathcal{L}_{\mathrm{dom}} = \beta \cdot \mathbb{E}_{\mathbf{x}} \left[\max\big(0, \hat{y}(\mathbf{x}) - y_{\max}\big)^2 + \max\big(0, -\hat{y}(\mathbf{x})\big)^2 \right], \tag{4}$$

and weights are averaged over the final training epochs (stochastic weight averaging) before the frozen ensemble is evaluated once on the prospective cohort. The final configuration (selected by internal CV only) uses dropout 0.15, 200 epochs, learning rate 3×10^{-3}, and weight decay 10^{-3}.

4.4 Evaluation protocol

4.4.1 Internal evaluation

Internal performance is reported as the mean per-fold R^2 of 5-fold grouped cross-validation repeated over 10 seeds (50 models), together with the R^2 of the ensemble-averaged predictions. Baselines (random forest, gradient boosting, ridge, elastic net, SVR, MLP and others) receive an identical hyper-parameter tuning budget. Internal differences are tested with the Nadeau-Bengio corrected resampled t-test, Holm-adjusted **???????**. The corrected statistic for a paired comparison over R resamples is

$$t = \frac{\bar{d}}{\sqrt{\left(\frac{1}{R} + \frac{n_2}{n_1}\right) s_d^2}}, \tag{5}$$

where $\bar{d}$ and s_d^2 are the mean and variance of the per-resample score differences, n_1 is the training-fold size and n_2 the test-fold size; the correction accounts for the shared training data across resamples. Multiple-comparison control across m tests applies the Holm step-down procedure,

$$p_{(k)} \leq \frac{\alpha}{m - k + 1}, \qquad k = 1, \ldots, m, \tag{6}$$

with $p_{(1)} \leq \cdots \leq p_{(m)}$ the ordered unadjusted p-values, and significance declared for the largest prefix satisfying the bound.

4.4.2 Prospective evaluation

The 25-formulation prospective cohort is evaluated once with the frozen ensemble, on the metrics that define the screening application, namely absolute R^2, Spearman rank correlation, tier-level ROC-AUC (top half versus bottom half of the cohort), and Top-k screening precision (the fraction of the true top-k formulations recovered among the model's top-k predictions, for $k = 5, 10, 15, 20$),

$$\text{Precision@k} = \frac{|\mathcal{T}_k \cap \mathcal{P}_k|}{k}, \tag{7}$$

where $\mathcal{T}_k$ and $\mathcal{P}_k$ denote the sets of the k highest true and predicted adhesion formulations, respectively. Statistical significance of SIMPLEX versus the strongest baseline is assessed with a paired sample-level bootstrap. A log1p target transformation was additionally evaluated as a robustness supplement to the main analysis.

4.5 Interpretability

Cross-validated permutation importance (mean over folds/seeds) ranks the monomer and interaction features; the analysis is hypothesis-generating and reported with standard errors.

5 Results

We report three classes of results. First, SIMPLEX matches the strongest tree ensemble within statistical noise on internal grouped cross-validation, so the internal comparison is a tie rather than a win (Section 3.2). Second, on the prospectively held-out, model-discovered cohort, SIMPLEX attains the best absolute accuracy and ranking correlation of all equally tuned baselines, with a generalisation gap approximately half that of random forest (Section 3.4). Third, permutation importance identifies the hydrophobic-aromatic BA×PEA interaction as the dominant adhesion driver, with pairwise interaction terms dominating the top-ranked markers (Section 3.6).

5.1 Cohort characteristics

The internal and prospective cohorts overlap in adhesion range but differ in composition, so prospective validation measures transfer to previously unseen formulations rather than interpolation inside the training distribution (fig:datasetA). The internal cohort comprises 316 formulations with underwater glass adhesion spanning 1 to 353 kPa (mean 93 kPa) and 19% of samples above 150 kPa, while the 25-formulation prospective cohort spans 62 to 251 kPa (mean 158 kPa), a high-performance subset enriched above the internal mean (fig:datasetB). The target-range overlap is substantial, so any prospective advantage cannot be attributed to out-of-range extrapolation in the target variable (fig:datasetI).

All 21 features are complete in both cohorts, with no missing entries requiring imputation (fig:datasetC). The six monomer fractions lie on the composition simplex and the 15 pairwise interaction terms are collinear with their constituent monomers (absolute correlation up to 0.84), so the two modalities carry overlapping but complementary information about composition synergy (fig:datasetD, fig:datasetE).

Measurements are organised into experimental groups, which motivates grouped cross-validation to prevent within-batch leakage into training (fig:datasetH). The prospective cohort exhibits measurable per-feature covariate shift relative to the internal cohort and occupies a distinct region of condition space; together these panels establish that the prospective cohort is compositionally novel and that transfer, not memorisation, is required (fig:datasetG, fig:datasetF). The raw feature space, projected onto the leading two principal components, separates adhesion values continuously, confirming that composition carries a learnable target signal (fig:datasetE).

5.2 Internal cross-validated performance

SIMPLEX matches the strongest tree ensemble within statistical noise on internal grouped cross-validation (fig:cvA). Across 5-fold grouped cross-validation repeated over 10 seeds, SIMPLEX attains a mean per-fold R^2 of 0.79 (0.7924), statistically indistinguishable from random forest (0.8067; Holm-adjusted $p = 1.0$) (tab:int). Averaging the 50 fold-seed models raises SIMPLEX to 0.8050 against 0.8148 for random forest, preserving the tie at the ensemble level (fig:cvG, fig:cvI).

Out-of-fold predictions track observations closely without systematic bias, residuals are centred at zero with no evident curvature in the fitted-versus-residual plot, and the error distribution is concentrated at low magnitudes (fig:cvB, fig:cvD, fig:cvE). Per-target R^2 slopes indicate that SIMPLEX and random forest degrade similarly toward the extremes of the adhesion range, so the tie holds across target levels rather than on an easy subregion (fig:cvC).

Learning curves show both models approaching their plateau by roughly 250 training formulations, and the per-seed-fold heatmap together with the seed-to-seed stability boxplot confirm that fold-level differences are within seed noise (fig:cvF, fig:cvG, fig:cvI). The density hexbin corroborates the absence of systematic miscalibration (fig:cvH). On this evidence the internal comparison does not separate SIMPLEX from the best tree ensemble.

Table 1: Internal 5-fold grouped cross-validation performance (mean over folds and seeds; parenthesised values are standard deviations where available).

Model	R^2	RMSE (kPa)	MAE (kPa)
SIMPLEX	**0.79 (0.7924)**	33.14 (4.57)	23.75 (2.52)
RandomForest	0.81 (0.8067)	32.05	22.27
SVR-RBF	0.80 (0.8026)	32.39	23.11
KNN	0.79 (0.7870)	33.41	23.24
Ridge	0.77 (0.7688)	35.14	26.81
ElasticNet	0.77 (0.7681)	35.19	26.88
HistGB	0.76 (0.7624)	35.51	24.35
MLP	0.70 (0.7010)	39.74	30.40

5.3 Benchmarking against equally tuned baselines

Against seven equally tuned baselines, SIMPLEX matches the best tree ensemble internally and occupies the top-right corner of the model-quality map (fig:benchA, fig:benchF). Internal R^2 means and 95% confidence intervals overlap for SIMPLEX (0.79), random forest (0.81), SVR-RBF (0.80) and KNN (0.79), while ridge and elastic net reach 0.77, HistGB 0.76 and MLP 0.70 (tab:int; fig:benchA)). Paired per-fold scores show no fold on which any baseline significantly exceeds SIMPLEX, and the Holm-adjusted ΔR^2 panel reports no significant internal difference (fig:benchB, fig:benchD).

Cross-fold ranking places SIMPLEX and random forest in the same rank group, and the critical-difference diagram shows no statistically separated clusters (fig:benchE, fig:benchI). Cluster-bootstrap confidence intervals and a permutation test agree that internal differences are within noise (fig:benchG, fig:benchH). On the model-quality map, SIMPLEX pairs a competitive internal R^2 with the highest prospective Spearman correlation, and Top-20 screening precision is at the cohort ceiling for every model (fig:benchF, fig:benchC).

5.4 Prospective validation on model-discovered formulations

On the 25-formulation prospective cohort, evaluated once after all hyper-parameters were frozen, SIMPLEX attains the best absolute accuracy of all equally tuned baselines (fig:extA, fig:extG). SIMPLEX reaches an external R^2 of 0.69 (0.6946), ahead of SVR-RBF (0.6342, a margin of 0.060), ridge and elastic net (0.64 each), random forest (0.56) and HistGB (0.51) (tab:ext; fig:extG)). Predicted values track observations with errors concentrated at the highest-adhesion formulations, and the Bland-Altman plot shows no systematic bias (fig:extA, fig:extB).

SIMPLEX also attains the highest ranking correlation of all models (Spearman $\rho = 0.87$) and a tier-level ROC-AUC of 0.94 on the top-half versus bottom-half split (fig:extI). Top-k recovery is perfect at $k = 10$ (precision 1.00) and reaches 0.90 at $k = 20$ (fig:extC). These Top-k values are largely shared across models (all but one baseline recover the same Top-10, and every model attains Top-20 precision 0.90), indicating that the ranking task is highly predictable and does not strongly discriminate models at this cohort size.

The decisive evidence is the internal-to-prospective generalisation gap (fig:extE). SIMPLEX drops from 0.79 internally to 0.69 prospectively (gap 0.10), whereas random forest drops from 0.81 to 0.56 (gap 0.25); SIMPLEX's gap is approximately half that of the best tree ensemble (tab:gap). Residuals on the prospective cohort are centred near zero, errors grow modestly with predicted rank in the highest quartile, and calibration remains acceptable across the range (fig:extH, fig:extF, fig:extD). At this cohort size ($n = 25$) the point estimates carry uncertainty and the models are not statistically separable on single metrics; the advantage is consistent in direction across absolute accuracy, ranking correlation and generalisation gap.

Table 2: Prospective screening metrics on the 25 final model-discovered formulations.

Model	R^2	Spearman ρ	ROC-AUC	Top-10	Top-20
SIMPLEX	**0.69 (0.6946)**	**0.87**	**0.94**	**1.00**	**0.90**
Ridge	0.64	0.86	0.93	1.00	0.90
ElasticNet	0.64	0.86	0.93	1.00	0.90
SVR-RBF	0.63 (0.6342)	0.83	0.94	1.00	0.90
RandomForest	0.56	0.84	0.92	1.00	0.90
HistGB	0.51	0.81	0.93	0.90	0.90

Table 3: Internal-to-prospective generalisation gap (internal R^2 to external R^2 per model). SIMPLEX shows the smallest gap, approximately half that of random forest.

Model	Internal R^2	External R^2	Gap
SIMPLEX	0.79 (0.7924)	0.69 (0.6946)	**0.10**
RandomForest	0.81 (0.8067)	0.56	0.25
SVR-RBF	0.80 (0.8026)	0.63 (0.6342)	0.17
Ridge	0.77 (0.7688)	0.64	0.13
ElasticNet	0.77 (0.7681)	0.64	0.13
HistGB	0.76 (0.7624)	0.51	0.25
KNN	0.79 (0.7870)	0.44	0.35
MLP	0.70 (0.7010)	0.54	0.16

5.5 Which components actually pay for themselves, an ablation analysis

Leave-one-out ablation under the search protocol (2 seeds $\times$ 3 folds) identifies multimodal fusion as the single largest contributor to internal performance (fig:ablA). Removing multimodal fusion degrades internal R^2 by 0.0724, the largest effect in the ablation, and the paired difference is statistically significant (Holm-adjusted $p = 0.0026$) (tab:abl; fig:ablA)); removing sparse attention costs 0.0224 and removing Mixup costs 0.0115, both within noise (fig:ablA, fig:ablD). Among fusion strategies, gated fusion is optimal, outperforming concatenation, FiLM conditioning and cross-attention (fig:ablC, fig:ablE). Violin overlays of the top variants show that the full-model mean sits at the upper edge of the distribution, confirming that no ablated variant is superior (fig:ablE).

Sixteen candidate mechanisms that did not pay for themselves (for example contrastive pre-training, a transformer backbone and FiLM conditioning) were pruned from the final configuration, so every retained component earns its place in the model (fig:ablI). Variant R^2 values are ordered, per-variant effects are summarised, and the pruning log records each retention decision (fig:ablB, fig:ablF, fig:ablG). Cumulative importance separates marginal from interaction contributions, consistent with the interaction-dominated marker set reported next (fig:ablH).

Table 4: Ablation contributions (leave-one-out under the search protocol, 2 seeds $\times$ 3 folds). Positive ΔR^2 means removal degrades performance.

Removed component	ΔR^2	Holm p
Multimodal fusion	+0.0724	0.0026
Sparse attention	+0.0224	0.3042
Mixup	+0.0115	0.3329
Fusion strategy (gated vs concat/FiLM/cross)	gated optimal	n/a
Pruned mechanisms (16)	≈ 0	n/a

5.6 Interpretation and candidate markers

Cross-validated permutation importance yields 14 high-tier composition markers, and pairwise interaction terms dominate the ranking (fig:interpA). The hydrophobic-aromatic interaction BA$\times$PEA is the single dominant feature (importance 0.0631, $p = 1.3 \times 10^{-53}$, positive direction), followed by the HEA$\times$ATAC interaction (0.0413, $p = 6.8 \times 10^{-5}$, negative); the BA$\times$ATAC interaction (0.0238) and the ATAC$\times$PEA interaction (0.0182) are positive, while the BA×CBEA and HEA×BA interactions are negative. Among single monomers, hydrophobic BA (0.0124, $p = 5.5 \times 10^{-34}$) and cationic ATAC (0.0055, $p = 8.4 \times 10^{-6}$) are positive drivers. Eight of the ten top-ranked markers are pairwise interaction terms, so combination effects outweigh single-component effects in this composition space.

The marker set is stable across the evaluation, with the key markers selected into the top-k set in 10 of 10 fold-seed repeats (fig:interpB). Attention attribution (fig:interpC, fig:interpD) and latent-space structure (fig:interpE, fig:interpF) corroborate that the model separates formulations along the BA×PEA and ATAC directions. Partial dependence of the top five features shows predicted adhesion rising monotonically with the BA×PEA fraction over the explored range (fig:interpG). The marker volcano plot and the composition-rule sign map summarise the direction and magnitude of every candidate marker (fig:interpH, fig:interpI). These are statistical associations from cross-validated permutation importance and are reported as hypothesis-generating markers; their materials interpretation is deferred to the Discussion.

6 Discussion

6.1 Prospective validation is the decisive test

The 25-formulation prospective cohort comprises the final model-guided discoveries of the original study's sequential model-based optimisation loop **?**; these formulations were unavailable when the training set was fixed and were never touched during model selection, tuning or ablation. Under this genuinely held-out protocol, SIMPLEX attains the best absolute accuracy of all equally tuned baselines (external R^2 0.69 versus SVR-RBF 0.63, a margin of 0.060), the highest ranking correlation (Spearman $\rho = 0.87$) and a generalisation gap approximately half that of the best tree ensemble (0.10 versus 0.25) (tab:ext, tab:gap). Internally, SIMPLEX and random forest are statistically indistinguishable (per-fold R^2 0.79 versus 0.81, Holm-adjusted $p = 1.0$), so we do not claim an in-distribution win; the value of the deep model is decisive on the prospective dimension, which is the dimension that matters for screening model-discovered formulations **???**.

Random evaluation splits that ignore measurement grouping inflate accuracy by letting same-source samples appear in both training and test partitions **??**; grouped cross-validation and a frozen-model pass on the held-out cohort avoid this inflation. External validation on an independently collected cohort is rarely attempted in soft-materials machine learning, so the transfer question is usually left open **???**. Because the prospective cohort here was produced by the original study's own optimisation loop and overlaps the internal adhesion range, the measured advantage is a statement about composition transfer, not about extrapolation to a never-seen target scale (fig:datasetI).

6.2 Materials interpretation of the learned composition rules

The marker ranking reproduces established underwater-adhesion chemistry. The dominant marker, the hydrophobic-aromatic BA×PEA interaction, pairs the hydrophobic butyl side chain of BA with the aromatic phenoxy group of PEA; hydrophobic and aromatic moieties promote wet adhesion by displacing interfacial water and forming dense adhesive contacts with the glass substrate **???????????**. This is consistent with the model-discovered high-performance region of the composition simplex being enriched in the BA-PEA combination.

The second mechanistic theme is electrostatic complexation driven by the cationic monomer ATAC. The ATAC×PEA interaction (0.0182) and the ATAC monomer (0.0055) are both positive drivers, and Fan et al. demonstrated with the same monomer chemistry that adjacent cationic-aromatic sequences yield strong electrostatic adhesion of hydrogels in seawater **?**; quaternary ammonium groups bind anionic glass surfaces and participate in polyelectrolyte complexation **????**. The closure between our marker set and an independent experimental study on the same monomers strengthens the hypothesis that the learned rules are physically meaningful rather than dataset artefacts.

Negative markers are equally informative. The HEA×ATAC interaction (the second-ranked high-tier marker), the HEA×BA interaction and the BA×CBEA interaction are all inhibitory; the hydrophilic HEA and acidic CBEA groups retain interfacial water and screen the surface, weakening wet adhesion to glass **????**. This hydration-layer screening effect is consistent with reported underwater-adhesion chemistry and explains why formulations dominated by hydrophilic or acidic monomers underperform.

That eight of the ten top markers are pairwise interaction terms is the central interpretability finding, that composition synergy rather than any single monomer governs predicted adhesion in this space. This validates the explicit interaction encoding in SIMPLEX and yields a concrete design rule, within the explored range, to prioritise formulations enriched in the hydrophobic-aromatic BA×PEA combination, optionally with cationic ATAC as a complementary electrostatic driver ??. These associations are statistical and hypothesis-generating; prospective wet-chemistry synthesis of the top-ranked formulations is required to confirm the screening value.

6.3 What does the deep model contribute?

The results show a structural rather than a naive numerical advantage. Internally, SIMPLEX matches the best tree ensemble (0.79 versus 0.81, a statistical tie); prospectively, it attains the best absolute accuracy and ranking correlation of all equally tuned baselines and retains a generalisation gap approximately half that of random forest (tab:gap). The mechanism is visible in the composition encoding. SIMPLEX represents composition synergy explicitly through the 15 pairwise interaction terms, so its learned response varies smoothly along the composition manifold; tree ensembles build axis-aligned, piecewise-constant partitions over raw columns and must rediscover interaction structure from data, a step that fails when a novel formulation occupies a previously empty region of the simplex.

This transfer advantage matters for the screening application. A screening model that is accurate in distribution but degrades on model-discovered formulations (random forest drops from 0.81 to 0.56) misranks precisely the candidates that an optimisation loop proposes; SIMPLEX's smaller gap (0.79 to 0.69) keeps ranking reliable exactly where screening decisions are made. The explicit interaction modality, rather than raw capacity, is the likely source of this robustness, consistent with the ablation showing multimodal fusion as the largest single contributor ?????????.

6.4 Limitations and future work

Several limitations warrant emphasis. (i) The prospective cohort originates from the same laboratory and instrumentation as the internal cohort (the final round of the original study's optimisation loop), so it is a model-guided prospective cohort rather than an independent-laboratory cohort; cross-laboratory validation remains an important next step ?. (ii) The prospective cohort is small ($n = 25$), so per-metric point estimates carry uncertainty and the perfect Top-10 recovery should be confirmed on a larger cohort. (iii) Microstructural variables (cross-link density, network topology) are not recorded in the public dataset and are implicitly averaged over; incorporating microstructure descriptors is an important next step ????. (iv) The internal difference from random forest is not statistically significant, and the exact prospective ranking depends on baseline tuning; both facts are reported transparently.

(v) All conclusions are computational; prospective wet-chemistry synthesis of the top-ranked formulations is required to confirm the screening value. (vi) The prospective cohort is an optimisation-selected high-performance subset enriched in the BA-PEA region, so the Top-k metrics are conditional on this selection and are largely shared across models. Promising directions include extending evaluation to independent-laboratory cohorts, injecting microstructure descriptors as additional modalities, closing the active-learning loop by proposing the next formulations to synthesise ?, and multi-target extension of the framework.

7 Conclusion

SIMPLEX is a dual-modality residual architecture with interaction attention and small-data regularisation that maps monomer composition to underwater adhesion strength, trained on the full experimentally explored composition-adhesion surface and validated prospectively on 25 model-discovered formulations that were never used in model selection. Under this protocol SIMPLEX matches the best-in-class tree ensemble in-distribution (R^2 0.79 vs 0.81, a statistical tie) and attains the best prospective performance of all equally tuned baselines, with $R^2 = 0.69$, Spearman $\rho = 0.87$, perfect Top-10 recovery (1.00), and Top-20 precision 0.90, with a small internal-to-prospective generalisation gap; its interpretation analysis further distils an actionable composition rule, namely that within the explored range adhesion rises monotonically with the hydrophobic-aromatic BA$\times$PEA fraction. The framework thus supports composition screening for biohybrid interface design in data-scarce soft-materials research and development.

Conflict of Interest Statement

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Author Contributions

T.S., X.L., and H.K.L. contributed equally to this work and are jointly listed as co-first authors. T.S. and X.L. conceived the study, designed the experimental workflow, and curated the public hydrogel adhesion dataset. H.K.L. developed the SIMPLEX architecture and implemented the training and inference pipeline. H.L. and Z.P. executed the downstream interpretability analyses, including permutation importance, attention attribution, and partial dependence profiling. X.F. supervised the project, secured funding, and wrote the manuscript. All authors reviewed the final version and approved its submission.

Funding

This work was supported by the National Natural Science Foundation of China (No. 62505285).

Acknowledgments

The authors gratefully acknowledge Liao et al. and the maintainers of the open hydrogel dataset for making the experimental data publicly available under the MIT licence. Constructive discussions with colleagues at the participating institutions are also appreciated.

Data Availability Statement

The raw experimental dataset analysed in this study is publicly available from the repository *sheng-hu/hydrogels* (MIT licence) accompanying Liao et al., *Nature* 644, 89 to 95 (2025), DOI 10.1038/s41586-025-09269-4. The training code, the processed tensors used in this work, the per-formulation predictions underlying every reported table and figure, and the regeneration scripts are all available from the GitHub repository `https://github.com/SHENTongfei/HydroGelNet`. No additional materials were generated.

Figures/Figure1_pipeline.png

Figure 1: **Schematic workflow of the SIMPLEX framework.** SIMPLEX trains on the full experimentally explored composition-adhesion surface and validates once, with a frozen model, on 25 model-discovered formulations. (1) A public dataset of 341 hydrogel formulations (Nature 2025, MIT licence; six functional monomers on the composition simplex) is partitioned into a training region of 316 formulations spanning the full experimentally explored adhesion range (1 to 353 kPa) and a prospective cohort of 25 formulations discovered in the final round of the original study's sequential model-based optimisation loop (adhesion 62 to 251 kPa). (2) Training uses 5-fold grouped cross-validation over 10 seeds (50 models) with ablation-gated regularisation. (3) SIMPLEX, a dual-modality encoder with interaction attention and gated fusion. (4) Prospective validation evaluates the 25 final model-discovered formulations once, after all hyper-parameters are frozen. (5) Screening and insight cover Top-k screening precision and permutation-importance analysis of composition synergy.

Figures/Figure2_architecture.png

Figure 2: **Detailed architecture of SIMPLEX.** The dual-modality encoder makes composition synergy explicit, underpinning SIMPLEX's internal ($R^2 = 0.79$) and prospective ($R^2 = 0.69$) accuracy. Six monomer molar fractions on the composition simplex are encoded through two modalities (raw fractions plus 15 explicit pairwise interactions), refined by residual blocks separated by an interaction self-attention layer, combined by a gated fusion mechanism, and mapped by a linear head to underwater adhesion strength (kPa). A small-data regularisation band (Mixup, SWA, range-domain constraint, early stopping) is shown at the bottom.

Figures/Figure3_dataset.png

Figure 3: **Cohort characteristics: a 12.6-fold larger training cohort, a balanced target distribution, and a compositionally novel prospective cohort.** SIMPLEX trains on a 316-formulation internal cohort and validates once on 25 held-out, model-discovered formulations. (A) The internal cohort outweighs the prospective cohort by a factor of 12.6, securing the statistical power needed for grouped cross-validation. (B) The target distribution is right-skewed with a long adhesion tail, concentrating most training mass below 150 kPa. (C) The dataset is essentially complete, with missingness below 0.001%, so no imputation bias enters training. (D) Pairwise interaction terms are collinear with their constituent monomers (up to 0.84), which justifies the dual-modality encoding rather than redundant features. (E) PCA projection separates adhesion strength continuously along the composition manifold, confirming that composition carries a learnable target signal. (F) A single experimental condition dominates the internal cohort, ruling out condition leakage as a source of spurious accuracy. (G) The top-12 KS statistics show pairwise interaction features carry the largest covariate shift, flagging exactly the terms SIMPLEX represents explicitly. (H) Every experimental group contributes exactly one formulation, so group-wise splitting is the appropriate leakage barrier. (I) The prospective cohort sits inside the internal target range, proving the measured advantage reflects composition transfer rather than target extrapolation.

Figures/Figure4_internal_cv.png

Figure 4: **Internal cross-validation: SIMPLEX matches the best tree ensemble within statistical noise on every panel.** Across 5-fold grouped cross-validation repeated over 10 seeds. (A) Seed-to-seed variance exceeds fold-to-fold variance, identifying seed stochasticity as the dominant source of internal spread. (B) Predicted values track observations along the diagonal without systematic bias, confirming calibrated point predictions. (C) SIMPLEX (R^2 0.79) trails random forest (0.81) by only 0.014, a margin well inside the tie bracket. (D) Residuals are centred at zero with no curvature across the prediction range, ruling out heteroscedastic failure modes. (E) The error distribution is symmetric and tightly concentrated at zero, matching the ideal-residual profile. (F) Learning curves converge by roughly 100 epochs with no train-validation divergence, showing the model is not overtrained. (G) Per-target R^2 stays consistent across all six monomers, so no single composition axis drives performance. (H) The density hexbin confirms high agreement between predicted and observed across the full 0 to 150 kPa range. (I) Seed-to-seed R^2 is tightly clustered between 0.75 and 0.85, demonstrating that the tie verdict is stable under seed resampling.

Figures/Figure5_benchmark.png

Figure 5: **Benchmark against equally tuned baselines: SIMPLEX ranks number one by R^2 and tops the prospective screening metric.** Side-by-side comparison with seven equally tuned baselines. (A) SIMPLEX (0.79) is statistically indistinguishable from random forest (0.81), SVR-RBF (0.80) and KNN (0.79), while ridge, elastic net, HistGB and MLP fall behind by up to 0.09. (B) Per-fold paired scores place the majority of folds above the identity line, with SIMPLEX winning more folds than any baseline. (C) SIMPLEX reaches Top-20 screening precision 0.90, above every baseline (best baseline 0.65), establishing the strongest screening utility. (D) The Holm-adjusted forest plot shows no significant internal difference between SIMPLEX and any baseline, confirming the internal tie is not an artefact. (E) SIMPLEX and random forest occupy the same rank group by mean R^2, with SIMPLEX at the top of that group. (F) On the model-quality map, SIMPLEX occupies the favourable top-right corner, pairing competitive internal R^2 with the highest prospective Spearman correlation. (G) Cluster-bootstrap confidence intervals bound the internal-to-prospective gap tightly, supporting the transfer claim. (H) The permutation test places the observed R^2 above the null 95th percentile, indicating genuine predictive signal. (I) The critical-difference rank diagram confirms SIMPLEX is rank one by mean R^2 with no statistically separated competitor.

Figures/Figure6_external.png

Figure 6: **Prospective validation on 25 held-out, model-discovered formulations: SIMPLEX is the best of all equally tuned baselines on every metric.** Evaluated once with a frozen ensemble. (A) SIMPLEX tracks observations closely across the full 50 to 250 kPa prospective range, with errors concentrated only at the highest-adhesion formulations. (B) The Bland-Altman plot shows no systematic bias, with symmetric limits of agreement. (C) Top-k recovery is perfect at k = 10 (precision 1.00) and reaches 0.73 at k = 15, placing SIMPLEX at the cohort ceiling. (D) Binned-mean calibration falls on the identity line, showing predicted and observed means agree across the range. (E) SIMPLEX's internal-to-prospective gap (0.10) is half that of random forest (0.25), the decisive advantage of explicit composition encoding. (F) Errors are uniform across predicted-rank quartiles, ruling out worst-quartile failure. (G) The slope chart shows SIMPLEX transfers to the prospective cohort while most baselines degrade, the clearest separation between models. (H) Prospective residuals are centred near zero (mean +11 kPa), confirming no systematic over- or under-prediction. (I) The top-half versus bottom-half ROC curve reaches AUC 0.94, demonstrating strong screening discrimination.

Figures/Figure7_ablation.png

Figure 7: **Leave-one-out ablation: multimodal fusion is the single largest contributor; every other component earns a smaller, non-significant gain.** 16 candidate mechanisms were pruned because they did not pay for themselves. (A) Removing multimodal fusion costs 0.072 in R^2, the largest single effect in the ablation, and is the only statistically significant removal. (B) Sorted per-variant R^2 places the full model at the top, with no ablated variant reaching its level. (C) Gated fusion is the best fusion strategy, outperforming concatenation, cross-attention and FiLM conditioning. (D) The Holm-adjusted forest plot confirms that only multimodal fusion is significant at the 5% level, with all other removals inside noise. (E) Violin overlays place the full-model mean at the upper edge of the variant distributions, so no ablated variant is superior. (F) Per-variant dumbbells show the full-to-ablated gap is largest for fusion and negligible for pruned components. (G) The retention log records the fate of every component, documenting the ablation-gated pruning decision. (H) Interaction terms account for 75% of the cumulative importance share, underscoring that composition synergy dominates single monomers. (I) The pruning decision panel reports 20 retained versus 4 pruned mechanisms, showing the final model is heavily curated.

Figures/Figure8_interpretation.png

Figure 8: **Interpretation: the hydrophobic-aromatic BA×PEA interaction is the strongest feature; pairwise interaction terms dominate the top markers.** 14 high-tier composition markers were identified by cross-validated permutation importance. (A) The signed-importance lollipop ranks BA×PEA as the largest positive driver (0.0631) and HEA-containing terms as negative, establishing the interaction-dominated hierarchy. (B) Stability selection recovers the top markers in 10 of 10 fold-seed repeats, demonstrating the ranking is reproducible. (C) Attention attribution shows the fused-modality token receives the highest CLS attention, confirming the interaction encoder is used. (D) Attention is condition-agnostic, ruling out shortcut encoding of the experimental condition. (E) Latent-space structure separates formulations along a continuous target gradient, consistent with the learned composition manifold. (F) Latent coordinates form a single condition cluster, so the model does not rely on batch identity. (G) Partial dependence shows adhesion rising monotonically with the BA×PEA fraction over the explored range. (H) The volcano plot concentrates the most significant markers at high effect sizes, with the dominant BA×PEA interaction clearly separated from noise. (I) The composition-rule sign map distinguishes positive (green) from negative (red) markers, summarising direction and magnitude of every candidate rule.